

Exam 3 Questions (8.D – 11.C)

CHEM 1211K | Fall 2025

This exam includes coverage of the quantum model of the atom, periodic trends, lattice energy and ionic bonding, fundamentals of covalent bonding, drawing Lewis structures, enthalpy calculations involving bond enthalpies, molecular geometry and VSEPR theory, polarity, and valence bond theory.

QUESTION 1

1. Use the dropdown menus below to indicate the numbers of angular and radial nodes in a **3p** atomic orbital.

Angular nodes: [☐ 0 ☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5]

Radial nodes: [☐ 0 ☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5]

2. Use the dropdown menus below to indicate the numbers of angular and radial nodes in a **3d** atomic orbital.

Angular nodes: [☐ 0 ☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5]

Radial nodes: [☐ 0 ☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5]

3. Use the dropdown menus below to indicate the numbers of angular and radial nodes in a **5s** atomic orbital.

Angular nodes: [☐ 0 ☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5]

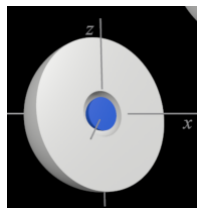
Radial nodes: [☐ 0 ☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5]

QUESTION 2

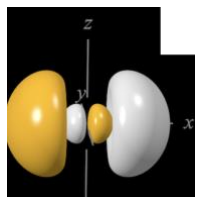
4. Consider the ground-state electron configuration of a neutral germanium (Ge) atom. Which of the following quantum numbers is associated with all the valence electrons in this atom?
- A. ☐ $n = 4$
 - B. ☐ $\ell = 1$
 - C. ☐ $m_\ell = -1$
 - D. ☐ $m_s = +1/2$
 - E. ☐ None of these; the quantum numbers of the valence electrons are all different.
5. Consider the ground-state electron configuration of a neutral sulfur (S) atom. Which of the following quantum numbers is associated with all the valence electrons in this atom?
- A. ☐ $\ell = 1$
 - B. ☐ $n = 3$
 - C. ☐ $m_\ell = -1$
 - D. ☐ $m_s = -1/2$
 - E. ☐ None of these; the quantum numbers of the valence electrons are all different.
6. Consider the ground-state electron configuration of a neutral tin (Sn) atom. Which of the following quantum numbers is associated with all the valence electrons in this atom?
- A. ☐ $\ell = 0$
 - B. ☐ $m_\ell = 0$
 - C. ☐ $m_s = +1/2$
 - D. ☐ $n = 5$
 - E. ☐ None of these; the quantum numbers of the valence electrons are all different.

QUESTION 3

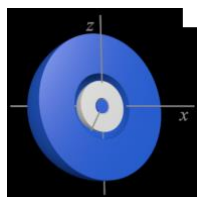
7. Which of the following images depicts an orbital with $\ell = 2$?



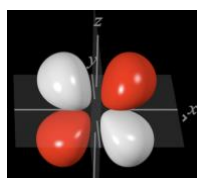
A. The orbital is spherical; this is a cross-sectional view.



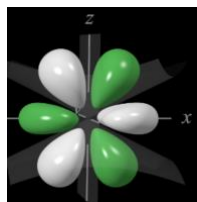
B.



C. The orbital is spherical; this is a cross-sectional view.

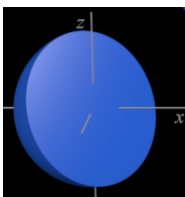


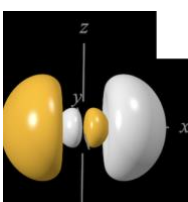
D.

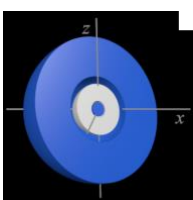


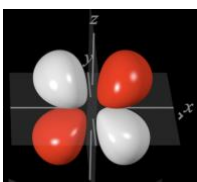
E.

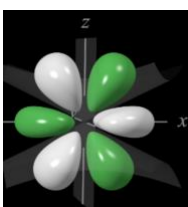
8. Which of the following images depicts an orbital with $\ell = 3$?

A.  The orbital is spherical; this is a cross-sectional view.

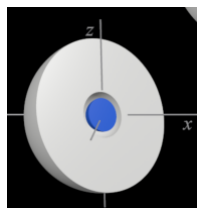
B. 

C.  The orbital is spherical; this is a cross-sectional view.

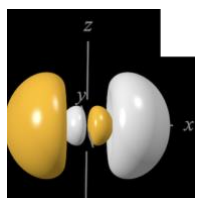
D. 

E. 

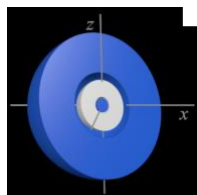
9. Which of the following images depicts an orbital with $\ell = 1$?



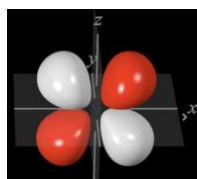
A. The orbital is spherical; this is a cross-sectional view.



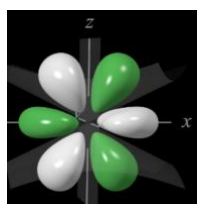
B.



C. The orbital is spherical; this is a cross-sectional view.



D.



E.

QUESTION 4

10. What is the total number of p electrons in a neutral atom of selenium (Se) in the ground state? Consider all shells. Enter your response as a numeral.
11. What is the total number of s electrons in a neutral atom of potassium (K) in the ground state? Consider all shells. Enter your response as a numeral.
12. What is the total number of s electrons in a neutral atom of barium (Ba) in the ground state? Consider all shells. Enter your response as a numeral.

QUESTION 5

13. Which of the following pairs of ions have identical electron configurations in the ground state?

- A. ☐ K^+ and Na^+
- B. ☐ O^{2-} and S^{2-}
- C. ☐ Al^{3+} and Cl^-
- D. ☐ H^+ and N^{3-}
- E. ☐ S^{2-} and Ca^{2+}

14. Which of the following pairs of ions have identical electron configurations in the ground state?

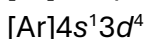
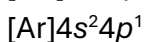
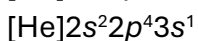
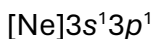
- A. ☐ S^{2-} and In^{3+}
- B. ☐ O^{2-} and F^-
- C. ☐ Li^+ and Na^+
- D. ☐ Ga^{3+} and Br^-
- E. ☐ H^+ and N^{3-}

15. Which of the following pairs of ions have identical electron configurations in the ground state?

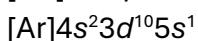
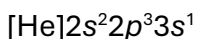
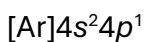
- A. ☐ Li^+ and Na^+
- B. ☐ O^{2-} and Be^{2+}
- C. ☐ Ca^{2+} and Cl^-
- D. ☐ Ga^{3+} and Br^-
- E. ☐ H^+ and N^{3-}

QUESTION 6

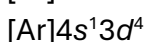
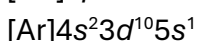
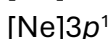
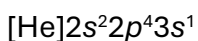
16. Each of the electron configurations below corresponds to an *excited state* of a neutral atom of one of the listed elements. Indicate the element corresponding to each electron configuration. Note that some elements may be used more than once and some not at all. Choices are *F*, *Mg*, *Ga*, *Sc*, *V*, and *O*.



17. Each of the electron configurations below corresponds to an *excited state* of a neutral atom of one of the listed elements. Indicate the element corresponding to each electron configuration. Note that some elements may be used more than once and some not at all. Choices are *F*, *Mg*, *Ga*, *Sc*, *V*, and *O*.



18. Each of the electron configurations below corresponds to an *excited state* of a neutral atom of one of the listed elements. Indicate the element corresponding to each electron configuration. Note that some elements may be used more than once and some not at all. Choices are *F*, *Mg*, *Ga*, *Na*, *V*, and *O*.



QUESTION 7

19. Rank the following ions from smallest (1) to largest (4).

Strontium cation | Selenide anion | Bromide anion | Rubidium cation

20. Rank the following ions from smallest (1) to largest (4).

Strontium cation | Yttrium(III) cation | Bromide anion | Rubidium cation

21. Rank the following ions from smallest (1) to largest (4).

Chloride anion | Calcium cation | Sulfide anion | Potassium cation

QUESTION 8

22. Absorption of a photon by a neutral strontium (Sr) atom causes excitation of a valence electron from the 5s subshell to the 5p subshell. Fill in the blanks below.

The ionization energy of the excited atom is [☐ higher than ☐ lower than ☐ equal to] the ionization energy of the ground-state atom because...

- [☐ the 5p subshell is higher in energy than the 5s subshell.
☐ the excited electron becomes unpaired.
☐ space is now available in the 5s subshell to accept an additional electron.
☐ the excited electron experiences less electron-electron repulsion.
☐ the excited electron feels greater effective nuclear charge.]

23. Absorption of a photon by a neutral strontium (Sr) atom causes excitation of a valence electron from the 5s subshell to the 5p subshell. Fill in the blanks below.

The electron affinity of the excited atom is [☐ higher in magnitude than ☐ lower in magnitude than ☐ equal to] the electron affinity of the ground-state atom because...

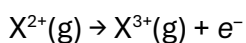
- [☐ the 5p subshell is higher in energy than the 5s subshell.
☐ the excited electron becomes unpaired.
☐ space is now available in the 5s subshell to accept an additional electron.
☐ the excited electron experiences less electron-electron repulsion.
☐ the excited electron feels greater effective nuclear charge.]

QUESTION 9

24. The electron affinity of magnesium cation (Mg^{2+}) is significantly higher in magnitude than the electron affinity of a neutral Mg atom. Which statement best explains why?
- A. ☐ The nuclear charge (Z) is greater in Mg^{2+} than Mg.
 - B. ☐ Because Mg^{2+} has the electron configuration of the noble gas Ne, it prefers not to accept an additional electron.
 - C. ☐ Electrons in Mg^{2+} experience less shielding than those in Mg.
 - D. ☐ The additional electron is forced to pair in Mg^- but can remain unpaired in Mg^+ .
 - E. ☐ None of these statements explain the observation.
25. The ionization energy of magnesium cation (Mg^{2+}) is significantly higher in magnitude than the ionization energy of a neutral Mg atom. Which statement best explains why?
- A. ☐ Electrons in Mg^{2+} experience greater effective nuclear charge than those in Mg.
 - B. ☐ The nuclear charge (Z) is greater in Mg^{2+} than Mg.
 - C. ☐ Because Mg^{2+} has the electron configuration of the noble gas Ne, it prefers not to lose an electron.
 - D. ☐ The lost electron is unpaired in Mg but paired in Mg^{2+} .
 - E. ☐ None of these statements explain the observation.

QUESTION 10

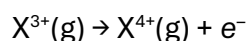
26. The ionization reaction below is significantly more difficult than the previous ionization. Fill in the blanks below to draw a plausible conclusion from this observation.



Neutral atoms of the element X contain [☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5] valence electron(s).

Of the following choices, the element X is most likely [☐ Na ☐ Mg ☐ Al ☐ Si ☐ P].

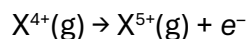
27. The ionization reaction below is significantly more difficult than the previous ionization.
Fill in the blanks below to draw a plausible conclusion from this observation.



Neutral atoms of the element X contain [☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5] valence electron(s).

Of the following choices, the element X is most likely [☐ Na ☐ Mg ☐ Al ☐ Si ☐ P].

28. The ionization reaction below is significantly more difficult than the previous ionization.
Fill in the blanks below to draw a plausible conclusion from this observation.



Neutral atoms of the element X contain [☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5] valence electron(s).

Of the following choices, the element X is most likely [☐ Na ☐ Mg ☐ Al ☐ Si ☐ P].

QUESTION 11

29. Which of the following iron salts has the greatest lattice energy?

- A. ☐ FeCl₂
- B. ☐ FeCl₃
- C. ☐ FeBr₂
- D. ☐ Fe₂S₃
- E. ☐ Fe₂O₃

30. Which of the following manganese salts has the greatest lattice energy?

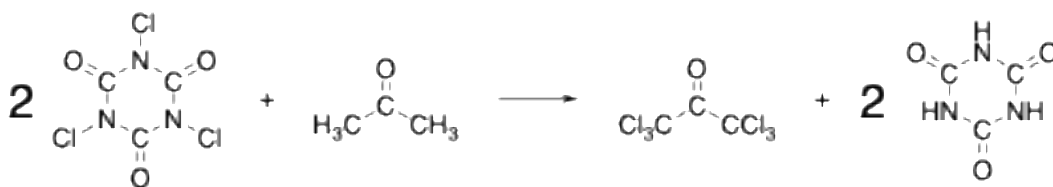
- A. ☐ MnF₂
- B. ☐ MnF₄
- C. ☐ MnCl₃
- D. ☐ Mn(NO₃)₂
- E. ☐ MnBr₄

31. Which of the following chromium salts has the greatest lattice energy?

- A. ☐ $\text{Cr}(\text{NO}_3)_2$
- B. ☐ $\text{Cr}(\text{NO}_3)_3$
- C. ☐ CrO_2
- D. ☐ CrF_2
- E. ☐ CrCl_3

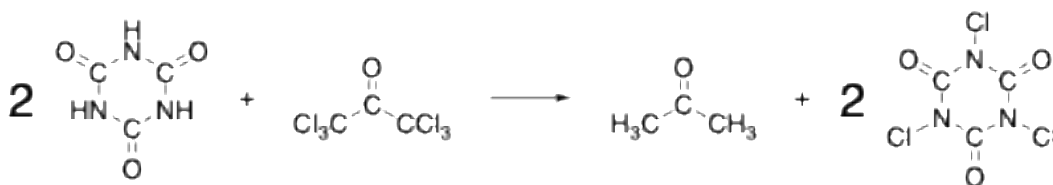
QUESTION 12

32. Using the table of bond enthalpies given below, determine $\Delta H^\circ_{\text{rxn}}$ for the reaction shown. Enter your response in kilojoules (kJ) to the nearest 1 kJ with appropriate sign.



Bond	Bond Enthalpy (kJ)
C=O	741
N—H	390
N—Cl	200
C—C	345
C—H	415
C—Cl	330

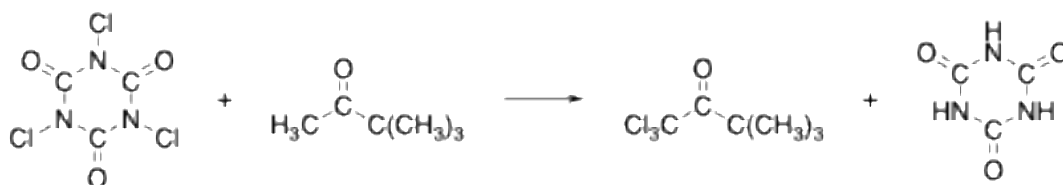
33. Using the table of bond enthalpies given below, determine $\Delta H^\circ_{\text{rxn}}$ for the reaction shown. Enter your response in kilojoules (kJ) to the nearest 1 kJ with appropriate sign.



Bond Bond Enthalpy (kJ)

C=O	741
N—H	390
N—Cl	200
C—C	345
C—H	415
C—Cl	330

34. Using the table of bond enthalpies given below, determine $\Delta H^\circ_{\text{rxn}}$ for the reaction shown. Enter your response in kilojoules (kJ) to the nearest 1 kJ with appropriate sign.

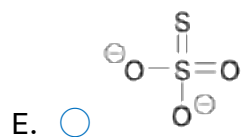
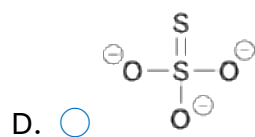
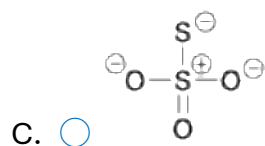
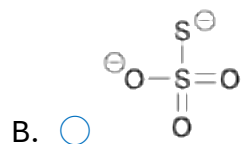
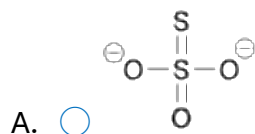


Bond Bond Enthalpy (kJ)

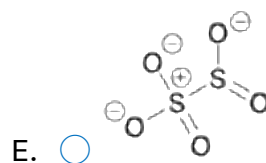
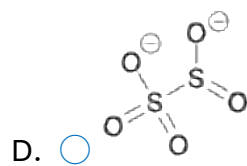
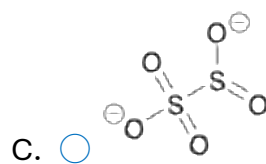
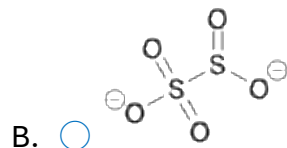
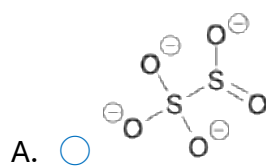
C=O	741
N—H	390
N—Cl	200
C—C	345
C—H	415
C—Cl	330

QUESTION 13

35. Which of the following structures is *NOT* a resonance form of the thiosulfate anion ($\text{S}_2\text{O}_3^{2-}$)?

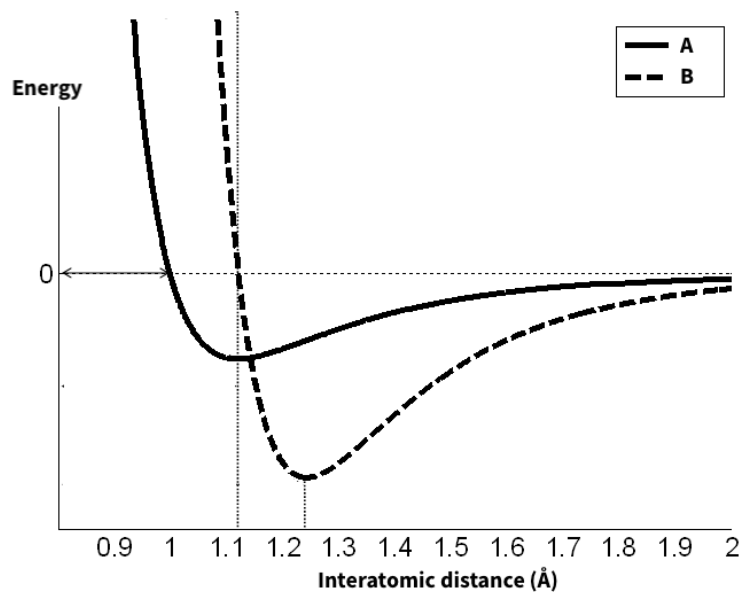


36. Which of the following structures is *NOT* a resonance form of the metabisulfite anion ($\text{S}_2\text{O}_5^{2-}$)?



QUESTION 14

37. The graph below shows how energy varies as atoms approach one another in two covalent bonds **A** and **B**. Fill in the blanks below.

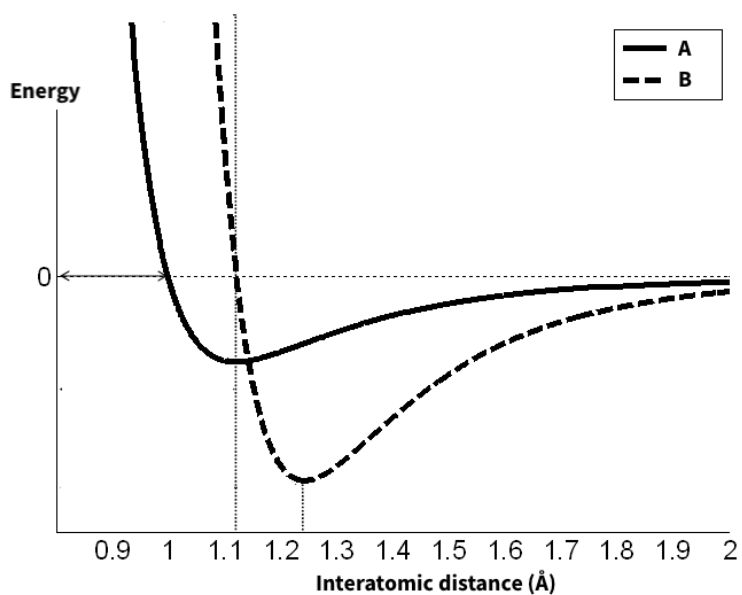


The stronger bond of the two is [☐ bond A ☐ bond B].

The shorter bond of the two is [☐ bond A ☐ bond B].

At infinite distance between the atoms, the energies of both bonds approach
[☐ negative infinity ☐ zero ☐ positive infinity].

38. The graph below shows how energy varies as atoms approach one another in two covalent bonds **A** and **B**. Fill in the blanks below.

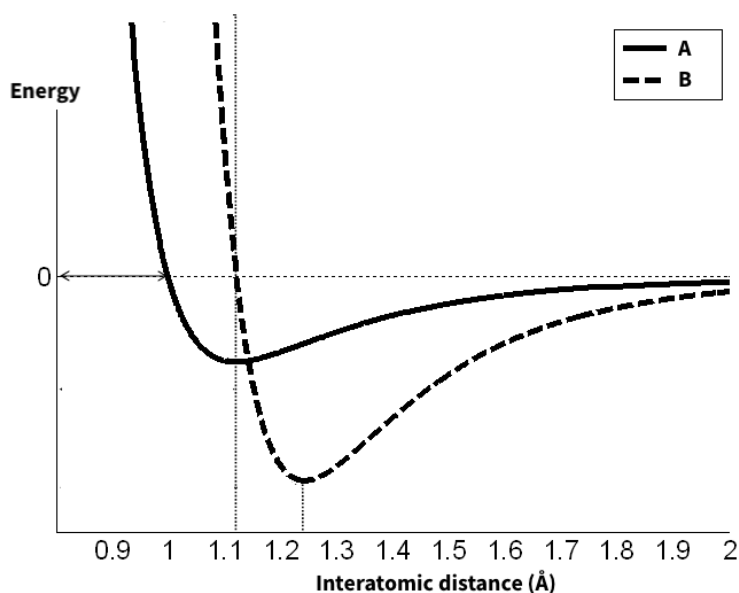


The weaker bond of the two is [☐ bond A ☐ bond B].

The longer bond of the two is [☐ bond A ☐ bond B].

At infinite distance between the atoms, the energies of both bonds approach
[☐ negative infinity ☐ zero ☐ positive infinity].

39. The graph below shows how energy varies as atoms approach one another in two covalent bonds **A** and **B**. Fill in the blanks below.



The weaker bond of the two is [☐ bond A ☐ bond B].

The shorter bond of the two is [☐ bond A ☐ bond B].

At infinite distance between the atoms, the energies of both bonds approach
[☐ negative infinity ☐ zero ☐ positive infinity].

QUESTION 15

40. Rank the following atoms from most electronegative (1) to least electronegative (4): S, As, P, Cl.

41. Rank the following atoms from most electronegative (1) to least electronegative (4): P, Sn, Ge, Si.

42. Rank the following atoms from most electronegative (1) to least electronegative (4): Si, N, F, Sn.